Scope of Work: Uninterruptible Power Supply (UPS) Maintenance Services

1.0 Objective
The purpose of this Scope of Work is to solicit proposals for a comprehensive preventive and corrective maintenance program for the Uninterruptible Power Supply (UPS) system identified below. The primary objective is to ensure the system's operational reliability and to maintain it in strict accordance with the original manufacturer's specifications and service recommendations.

2.0 Equipment Under Maintenance

- **Unit Type**: Uninterruptible Power Supply (UPS)
- **Manufacturer**: APC by Schneider Electric
- **Model**: MGE Galaxy 5000
- **Serial Number**: 3AER28004001
- **Power Rating**: 75kVA
- **Electrical Configuration**: 480V - 208Y/120V, 3-Phase, 4-Wire
- **Power Source**: Fed from panel CSBD4SE1 (room 1683)
- **Location**: 4735 E. Marginal Way S. Seattle, WA. 1st floor room 1683

3.0 Scope of Work
The selected contractor shall provide all necessary labor, tools, materials, and expertise to perform a full maintenance service on the UPS and associated battery systems. All procedures must align with the official service manual for the APC MGE Galaxy 5000 model.

3.1 Initial System Baselining and Remediation
It is understood that the UPS system has not been maintained in accordance with the manufacturer's recommended service intervals. Therefore, the contractor's first service visit shall be to establish a new operational baseline for the system. This initial service must, at a minimum, encompass all tasks outlined in both the Annual Preventive Maintenance (Section 3.2) and Semi-Annual Battery System Maintenance (Section 3.3) below.

During this baselining service, the contractor is required to perform a comprehensive diagnostic evaluation to identify any existing faults, degraded components, or other deficiencies resulting from the extended operational period without proper maintenance.

Any critical issues discovered must be immediately reported to the project lead, and a detailed quotation for necessary remedial work must be provided. Following the completion of this initial baseline service and any approved corrective actions, the system will be considered ready for the standard ongoing maintenance schedule.

3.2 Annual Preventive Maintenance
At a minimum, the following tasks shall be performed annually after the initial baseline is established:

- **Visual Inspection: Check for any signs of overheating, loose connections, corrosion, or physical damage.**
- **Cleaning: Thoroughly clean and vacuum the interior and exterior of the UPS and battery cabinets to ensure proper cooling and operation.**
- **Thermal Scan: Perform an infrared thermal scan of all electrical connections to identify potential hot spots.**
- **Parameter Verification: Measure and log all operational readings, including input/output voltages, currents, and frequency. Compare readings to manufacturer specifications.**
- **Functional Test: Perform a full operational test, including a managed transfer to bypass mode and back to normal online operation to verify the functionality of the static switch.**
- **Sub-assembly Inspection: Inspect and test major components including the rectifier, inverter, static switch, and battery charger.**
- **Connection Torque: Check and tighten all internal power connections as required.**
- **Filter Replacement: Inspect all cooling fans for proper operation and replace air filters.**

3.3 Semi-Annual Battery System Maintenance

At a minimum, the following battery maintenance tasks shall be performed every six (6) months after the initial baseline is established:

- **Visual Inspection: Inspect all battery jars/blocks for leaks, cracks, swelling, or corrosion.**
- **Connection Integrity: Clean all battery terminals and posts. Check and torque all inter-cell and terminal connections to manufacturer specifications.**
- **Health Checks: Measure and record the voltage and impedance of each individual battery cell/jar to trend battery health and identify failing units.**
- **Environment Check: Record the ambient temperature and ensure it is within the recommended range for optimal battery life.**

3.4 Lifecycle Component Management

The contractor is required to identify and report on critical components that have a limited service life, as per the manufacturer's guidelines for the MGE Galaxy 5000. This includes, but is not limited to:

- **DC Capacitors**
- **AC Capacitors**
- **Cooling Fans**

The service report must include the age/status of these components and provide a quote for their proactive replacement based on the manufacturer's recommended service interval.

4.0 Contractor Qualifications

- **Bidding contractors must be an authorized service provider for APC / Schneider Electric UPS systems.**
- **All technicians assigned to perform work must be factory-trained and certified on the MGE Galaxy 5000 series UPS. Proof of certification may be required.**
- **The contractor must provide a certificate of insurance.**

5.0 Deliverables
Following each service visit (both preventive and corrective), the contractor must provide a comprehensive electronic Field Service Report within five (5) business days. This report must include:

- **A detailed summary of all work performed.**
- **A log of all measurements and operational parameters recorded during the visit.**
- **The results of all functional and battery tests.**
- **A detailed list of any deficiencies found, along with clear recommendations and a quotation for corrective action.**
- **An overall assessment of the system's health and reliability.**